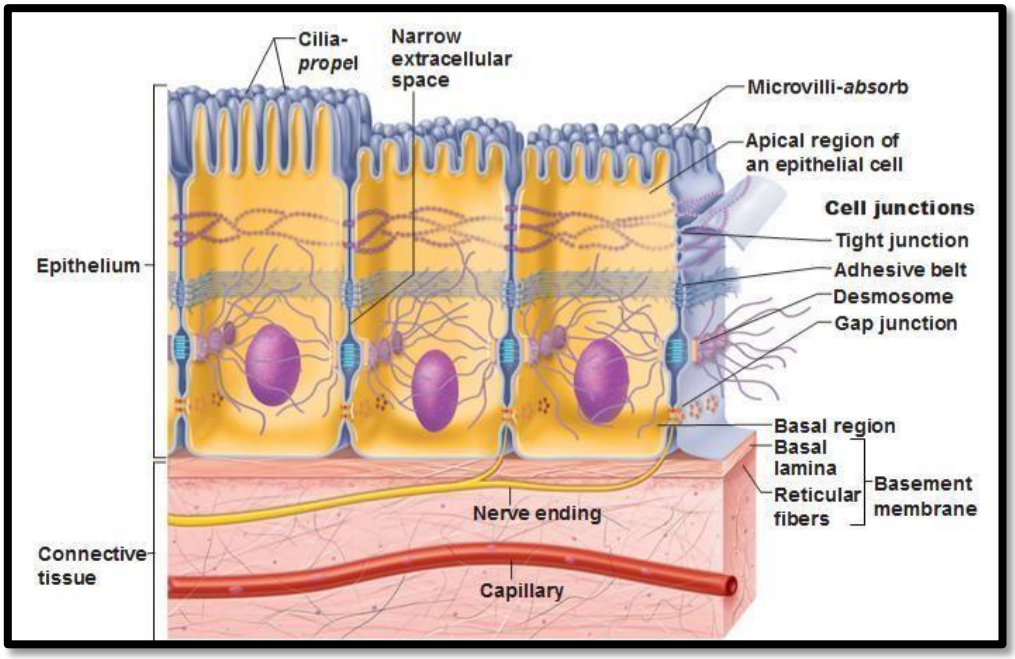
Epithelium

Epithelium Tissue

Epithelium

- Epithelium is one of the four primary **or** basic body tissues.
- It is formed of group of adherent cells with minimal **or** no intercellular substance.
- **General Characters of Epithelium**:
  1. Epithelial cells are **close together** and separated by a **little (or no) intercellular substance**.
  2. **Cell junctions** are present between its cells to help their fixation.
  3. Epithelium is separated from the underlying C.T. by the **basement membrane**.
  4. It has **no blood vessels** (not vascularized).
  5. The cells receive their nutrition by **diffusion** from underlying connective tissue.
  6. It has **no lymph vessels**.
  7. It has **numerous nerve endings**.
  8. Its cells have a **high power to regenerate** after degeneration.
  9. Its cells **exhibit polarity**.



- **Classification of epithelium**:

A) According to its functions, the epithelium is classified into:

1. Surface epithelium (membrane epithelium)
2. Glandular epithelium
3. Myoepithelium
4. Neuroepithelium

1. Surface Epithelium:

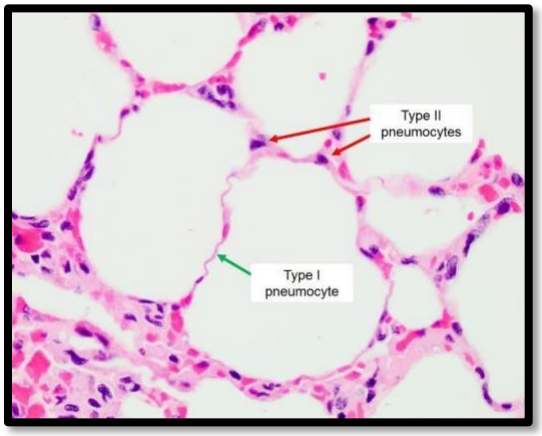
- The **lining** epithelium of **tubes** and **cavities** and the **covering** epithelium of **body surface**.
- Arranged to form **one continuous layer** or **layers** resting on a basement membrane.
- **They are classified according to the number of layers into**:

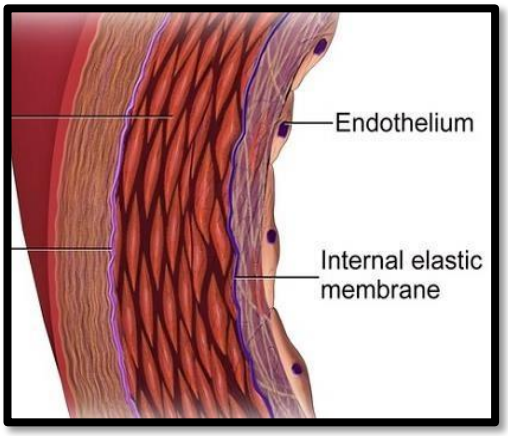
A. Simple Epithelium:

- It is a surface epithelium in which the cells are arranged in **one layer**.
- **According to the shape of its cells, it is classified into**:

Simple Squamous Epithelium:

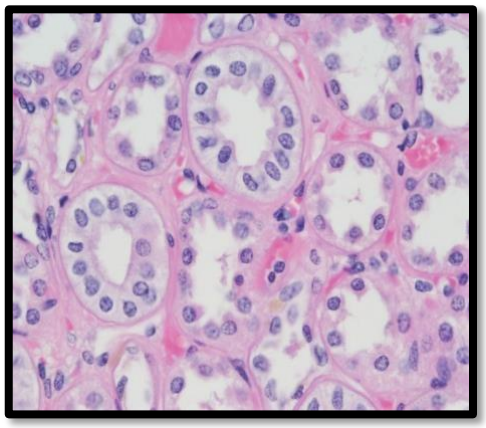
- **Shape**: flat cells with flat nuclei
- **Sites**:
  1. Lung alveoli.
  2. Endothelium of blood vessels.
  3. Serous cavities; pleura, pericardium and peritoneum (mesothelium).
- **Functions**: provides a smooth and thin surface to facilitate filtration and diffusion of water, electrolytes, and transport of gases.





Simple Cuboidal Epithelium:

- **Shape**: cubical in shape with central and rounded nuclei
- **Sites**:
  1. Thyroid follicles.
  2. Some kidney tubules.
  3. Ovarian surface.
- **Functions**: mainly absorption and secretion.



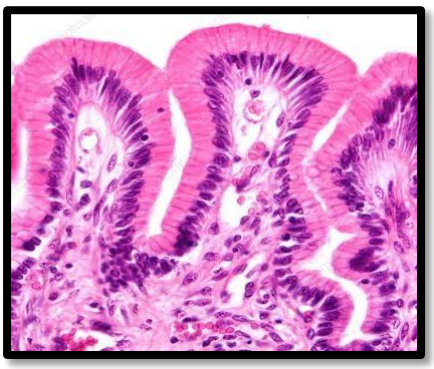
Simple Columnar Epithelium:

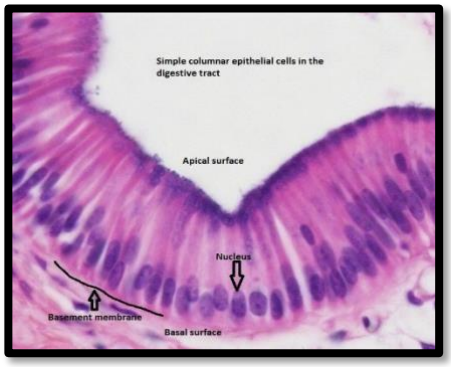
- **Shape**: tall cylindrical with basal and oval nuclei.
- **Types**:
  1. **Without brush border** e.g. epithelium of stomach.
  2. **With brush border** that seen by EM and may be:
     - **A. Cilia** e.g. epithelium of uterus and fallopian tubes.
     - **B. Microvilli** e.g. epithelium of small intestine.
- **Functions**: the function differs according to its site:

a. **Secretory** e.g. epithelium of the stomach.

b. **Absorptive** e.g. epithelium of small intestine.

c. **Movement** of ova and sperms in the uterus and fallopian tubes (ciliated epithelium).





Pseudostratified Columnar Epithelium:

- **Shape**:

✓ It is formed of one sheet of cells which rest on basement membrane.

✓ Some cells are tall with apical nuclei and others are short with basal nuclei.

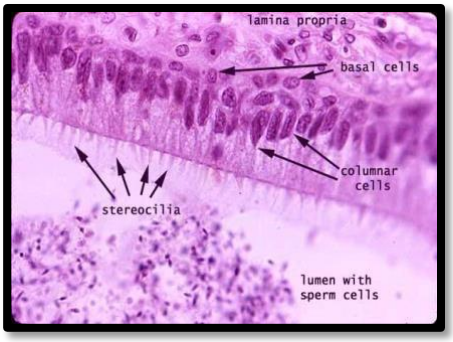
- **Types and sites**:

a. Non ciliated pseudostratified columnar epithelium: epithelium of large ducts of glands

b. Pseudostratified columnar ciliated epithelium: epithelium of trachea.

c. Pseudostratified columnar epithelium with stereocilia: epithelium of epididymis.

- **Functions**: protective and others according its site.



B. **Stratified Epithelium**:

- It is one of the surface epithelium in which the cells are arranged in **several layers**.

✓ **The basal layer** is usually formed of columnar cells.

✓ **The middle layers** are mostly formed of polyhedral cells.

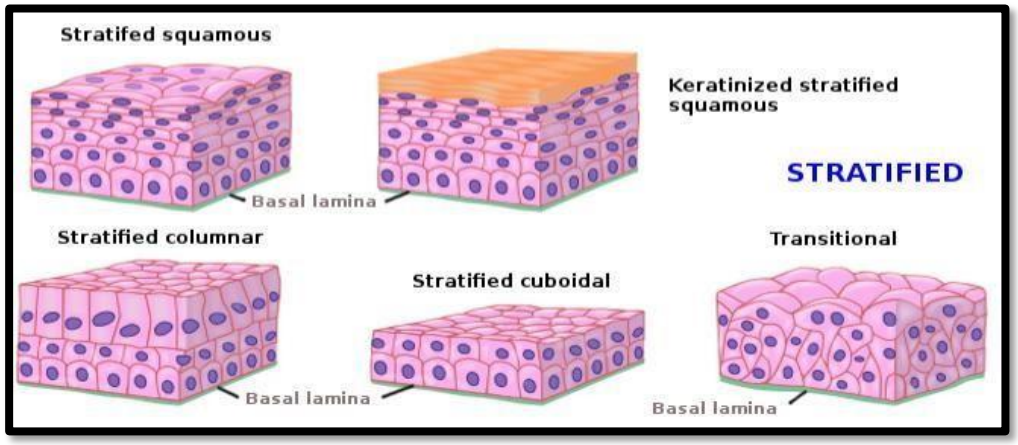
✓ **The superficial layers** the cells have different shapes.

- **Functions**:

✓ All types of stratified epithelium perform **protective** functions.

✓ Usually found in the surfaces subjected to **frictional forces** and **other potential hazards**.

- **According to the shape of cells in the superficial layers, it is classified into**:



Stratified Squamous Epithelium:

- **It is of two types**:

A. **Non-keratinized stratified squamous epithelium:**

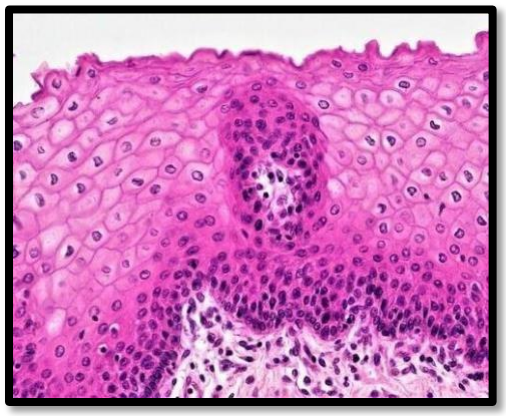
✓ The superficial cells are squamous and nucleated (**no keratin**).

✓ It lines the wet surfaces as **mouth**, **oral pharynx**, **esophagus**, **anal canal**, and **vagina**.

B. **Keratinized stratified squamous epithelium:**

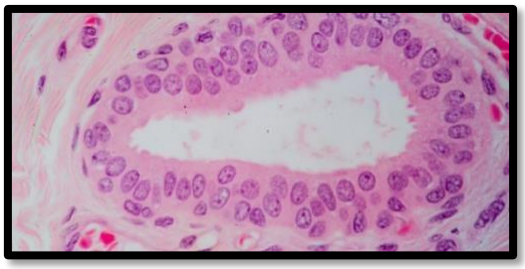
✓ The superficial layers are apoptotic cells, lose their nuclei & cytoplasm is replaced with **keratin**.

✓ It covers the dry surfaces as epidermis of the **skin**.



Stratified cuboidal epithelium:

- The superficial cells are cuboidal.
- It lines large ducts of exocrine glands as salivary and sweat glands.

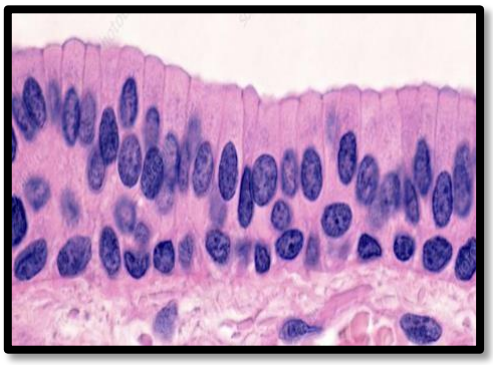


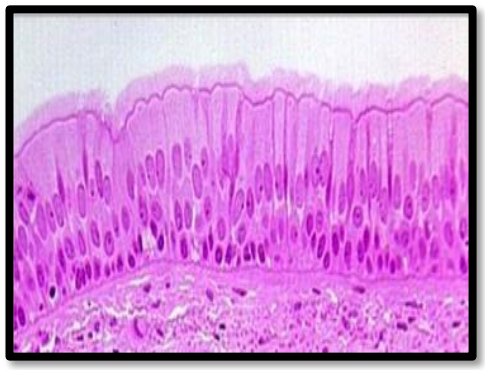
Stratified columnar epithelium:

- The superficial cells are columnar.
- **It is of two types**:

  A. **Non ciliated stratified columnar epithelium:** e.g. large ducts of exocrine glands, and one part of male urethra (penile part).

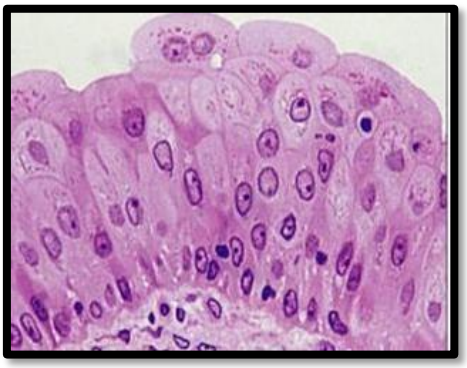
  B. Ciliated stratified columnar epithelium: e.g. fetal esophagus.





Transitional Epithelium:

- The superficial cells are **cuboidal** and **may be binucleated.**
- It is exclusively found in **urinary passages** including renal calyces, renal pelvis, ureter, urinary bladder and first part of urethra.
- It is specialized to accommodate greater stretch so the number of its layers is **changeable**; nearly 4-8 layers in the empty bladder and become 2-3 when the bladder is full.



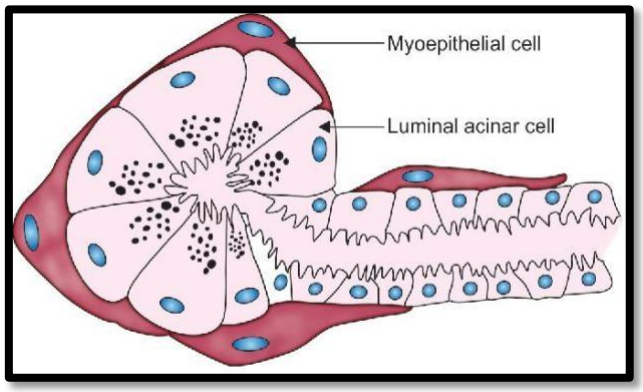
2. Myoepithelium:

Definition: Epithelial cells which have the ability to contract like the **muscle**.

Cytoplasm: These epithelial cells contain **myofilaments**.

Site: They surround the secretory exocrine acini, usually located **between the secretory cells and their basal lamina** e.g. salivary and sweat glands.

Function: Help **releasing secretory products** to the lumen of the acini.

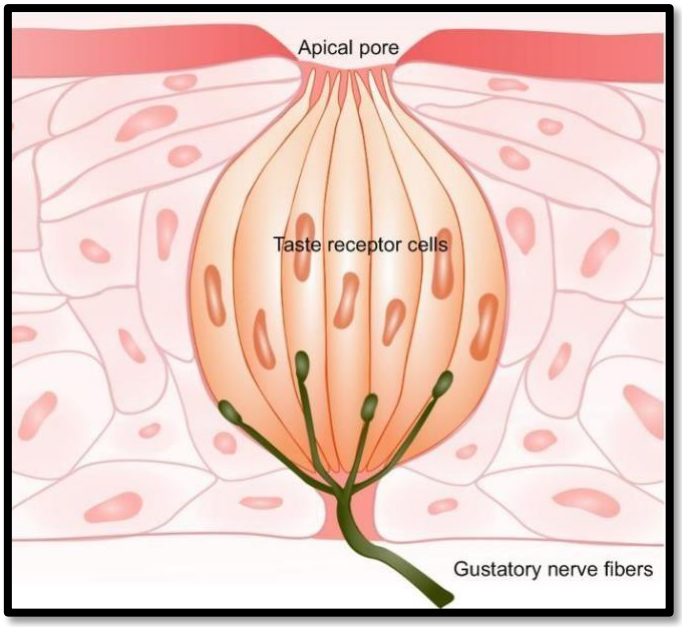


3. Neuroepithelium:

Definition: These are modified epithelial cells for **perception of nerve stimuli**.

Site:

1. **Tongue**: taste buds.
2. **Inner ear**: organ of corti (hearing), macula and crista ampularis (vestibular apparatus).
3. **Nose**: olfactory mucosa



4. Glandular Epithelium:

- **Definition**: is composed of group of epithelial cells, which are specialized for **secretion**.
- **Development**:

✓ The gland is formed by invagination of surface epithelium which either:

1 Remains **connected to the surface epithelium** with a duct forming an exocrine gland.

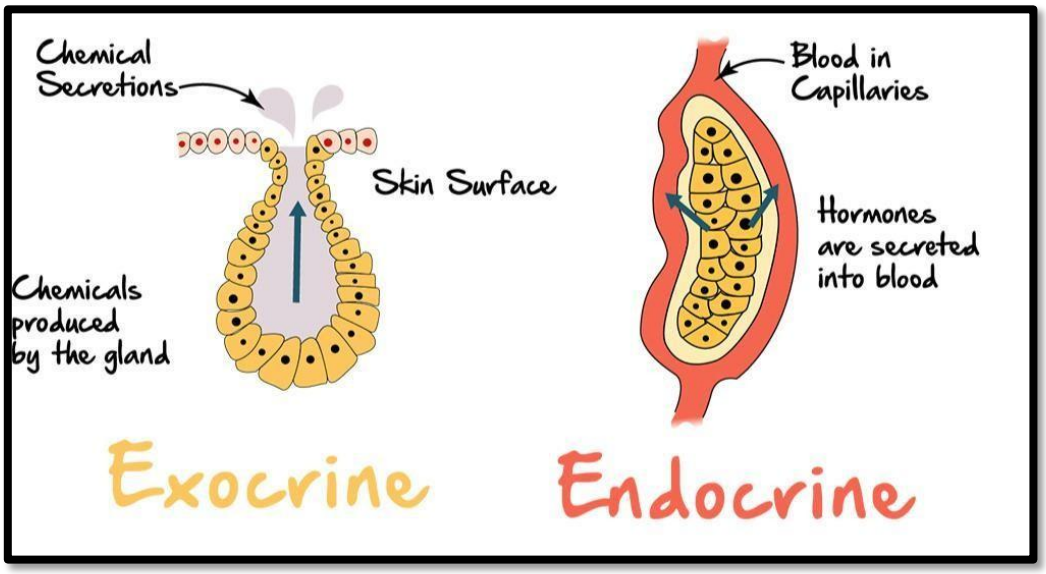
2 **Not connected to the surface** with a duct forming an endocrine gland.

- **Types**: **classified into 3 types according to presence or absence of duct system**:

A. **Exocrine glands**: glands that secrete their products to either external or internal surface of the body **through a duct** e.g. salivary and sweat glands.

B. Endocrine glands: glands that release their products (hormones) **directly in blood stream** as they lack duct system e.g. pituitary, thyroid, and suprarenal glands.

C. Mixed glands: glands that contain **secretory acini** forming an exocrine portion and **cells secreting hormones** forming an endocrine portion e.g. pancreas.



EXOCRINE GLANDS:

- **The classification of exocrine glands is based on different criteria**:

I. **Nature of Secretion**:

A. Mucous glands: Secrete mucin that forms viscous mucus e.g. goblet cells.

B. Serous (watery) glands: Secrete protein-rich watery fluid e.g. parotid gland.

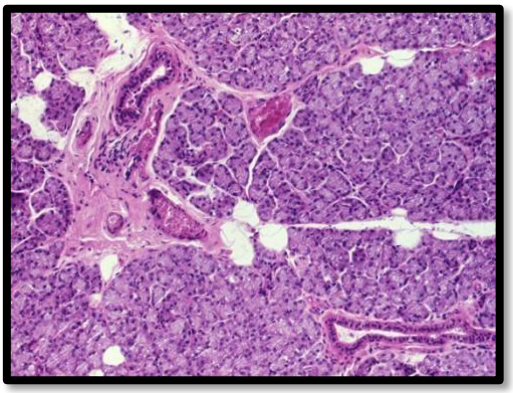
C. **Seromucous glands** (**mixed gland**): Secrete both serous and mucus secretions e.g. submandibular gland.

D. Glands with special secretions:

a) Fatty: e.g. sebaceous gland.

b) Waxy: e.g. glands of external ear (ceruminous glands).

c) Cellular: e.g. spermatozoa (testis) and ova (ovary).

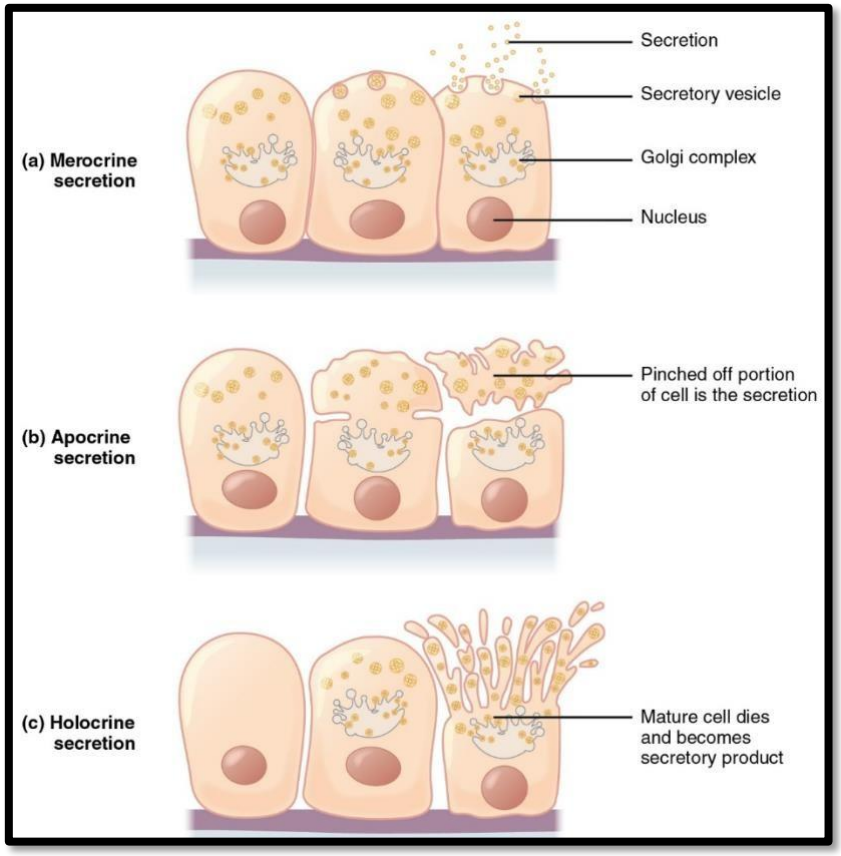


II. Mechanism (Mode) of Secretion:

A. **Merocrine glands**: release their secretion by **exocytosis** without any cellular changes e.g. **parotid gland**.

B. **Apocrine glands**: the **apex of the cell is released** with the secretion e.g. **mammary gland**.

C. **Holocrine glands**: secretory granules accumulate until the cell becomes filled (bloated) with the secretory products; then, **the cell ruptures** to release its contents e.g. **sebaceous gland**.



III. Number of Cells:

A. **Unicellular Glands**:

Goblet cell: secretory single cells distributed among non-secretory cells.

- **Shape**: is goblet (cup like):
  - ✓ Its **apex** is expanded with secretory granules.
  - ✓ Its **base** is narrow containing flattened nucleus and some organelles.
- **Staining**: **H & E** stained sections, the mucin is dissolved and the cells appear vacuolated. Therefore, a special stain e.g. **PAS** is used to clarify these cells.
- **Site**: in the epithelial lining of respiratory tract and intestine.
- **Function**: secretion of **mucus**.

B. **Multicellular Glands:**

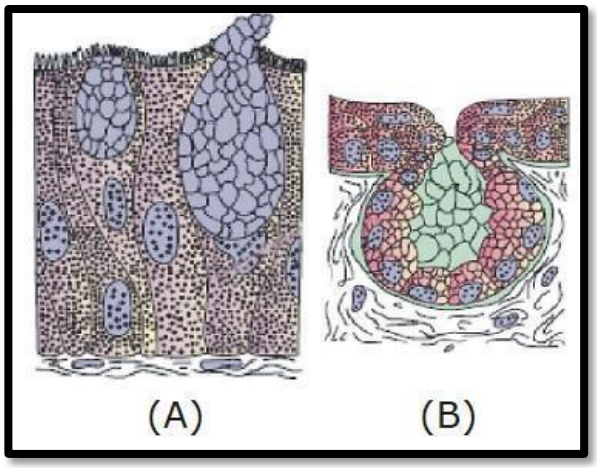
- These glands consist of **more than one cell**.
- **The exocrine multicellular glands are further classified according to the shape of secretory or conducting portions**:

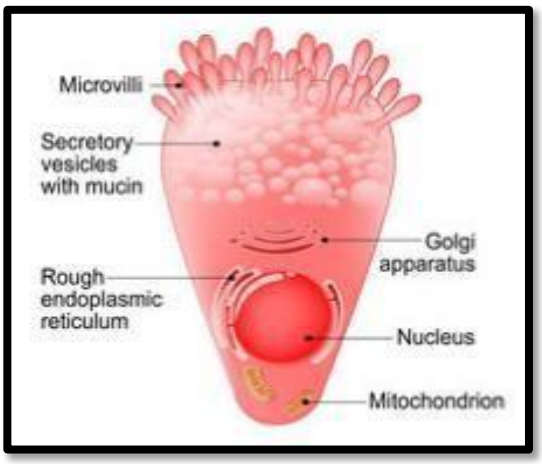
A) According to the shape of the secretory portion:

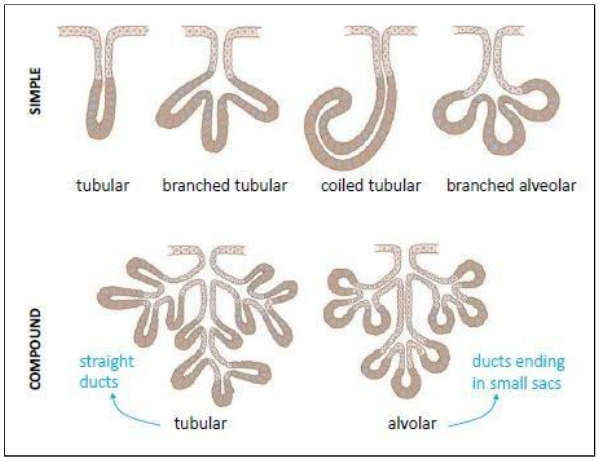
1. **Tubular glands**: the secretory cells form a tube-shaped lumen
2. **Acinar (grape) or alveolar (sac) glands**: The secretory cells are arranged around a roughly rounded or flask shaped lumen.
3. **Tubulo-alveolar glands**: the secretory part is a tube that ends with a sac like dilatation.

B) According to the shape of the duct portion:

1. **Simple glands**: glands that have a single unbranched duct.
2. **Compound glands:** glands that have branched ducts.



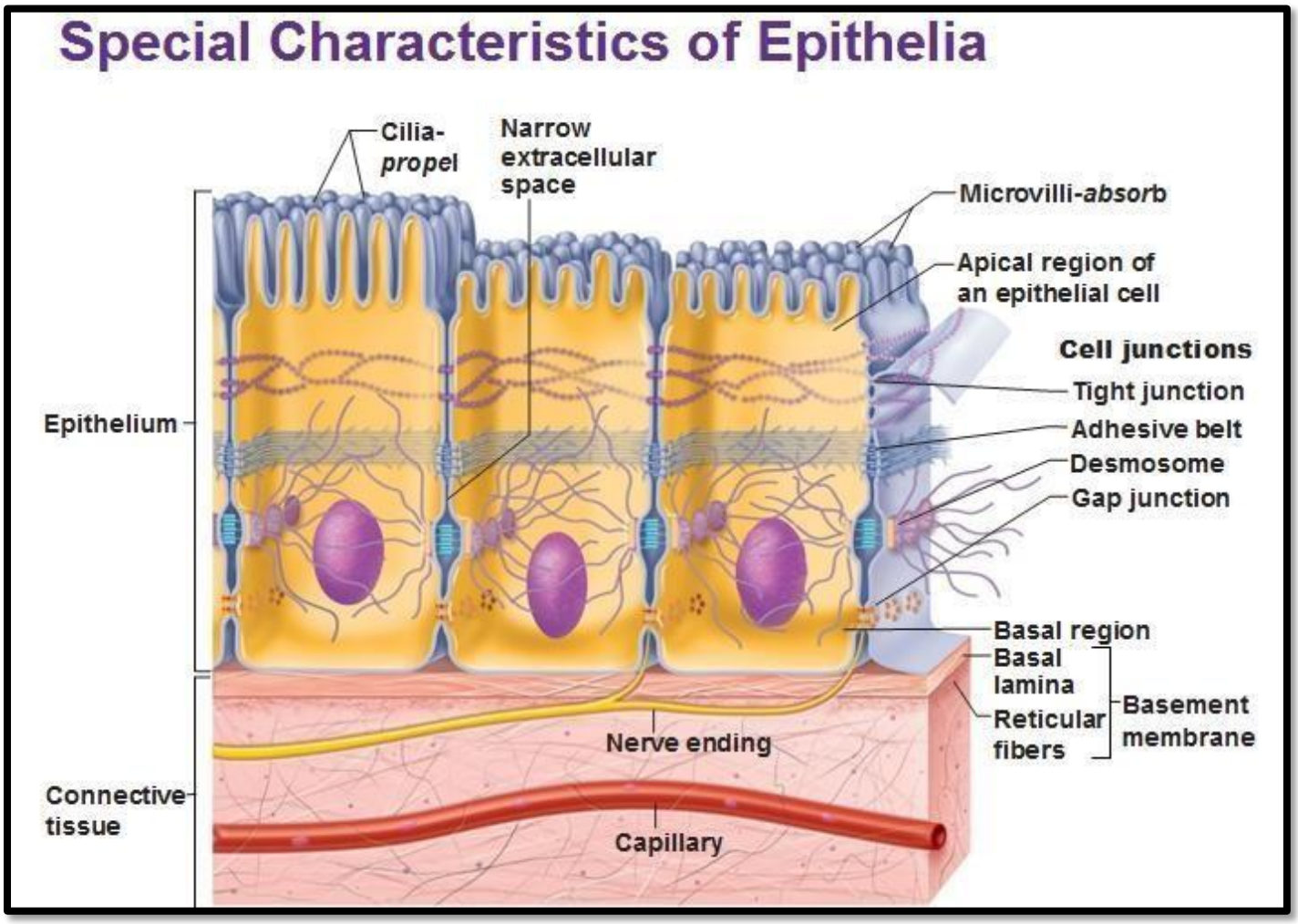




Epithelial Polarity

Cell membrane specializations

- **Definition**: epithelial cells are called polarized cells as we can distinguish their different surfaces with specific structural modifications in their cell membrane; apical, basal, and lateral domains.



- **Types**:

1) **Apical Surface Specializations**: The specific structures found on the apical surface (the free surface facing the lumen or external environment) include:

   a) Microvilli

   b) Cilia

   c) Flagella

   d) Stereocilia

A) Microvilli:

- **Definition**: They are **finger- like projections** from the apical surface of cell membrane
- **Site**: epithelial cells lining small intestine and proximal convoluted tubules of kidney
- **LM**: they appear as acidophilic striated **or** brush border
- **EM**:

✓ In longitudinal sections, appear as cylindrical processes containing **actin filaments**

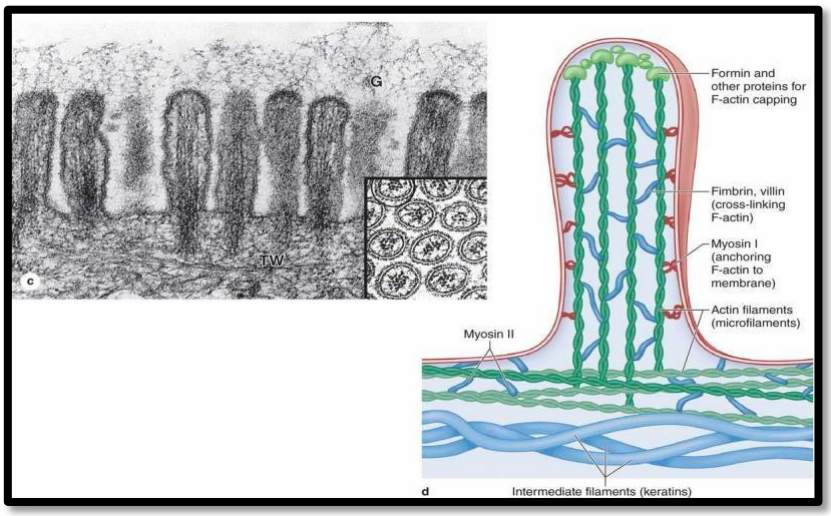
✓ These actin filaments are surrounded by the cell membrane.

✓ The tips of actin filaments are attached to the covering cell membrane while the bases of the filaments are attached to the **terminal web**.

✓ The terminal web is formed of transversely arranged **actin and intermediate filaments** at superficial zone of the cell cytoplasm.

✓ The covering cell membrane has a **glycocalyx**.

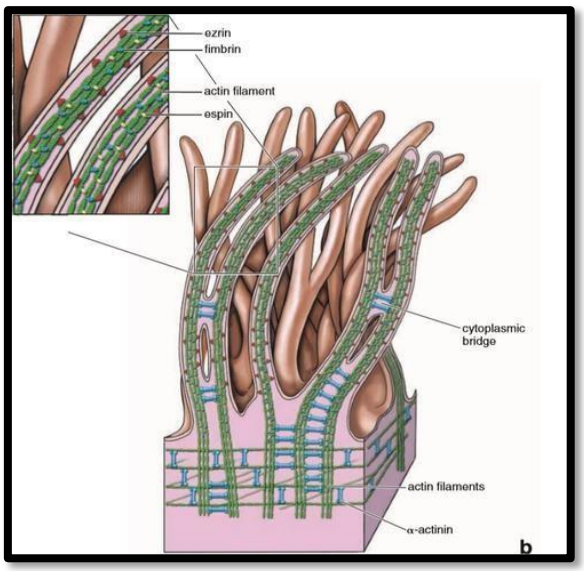
- **Function**: They increase the surface area for absorption by about 15-30 folds.



B) Stereocilia:

- **Definition**: They are long microvilli but non-motile.
- **Site**: epithelial cells lining epididymis and inner ear receptors.
- **LM**: they appear as striated border
- **EM**: like microvilli. (**MCQ**)
- **Function**:

1. **Absorptive** e.g. epidydimis
2. **Receptors** e.g. those of hair cells in ear



2) **Lateral surfaces specializations** (**Cell Junctions**):

- **Definition**: Cell membrane specializations found on the lateral surfaces of epithelial cells (between adjacent epithelial cells).
- **Function**:

1. Provide strong attachment to the adjacent cells.
2. Prevent passages of materials through the intercellular space.
3. Other functions: according to the junction type and cell location.

- **Classification**:

1 According to the distance between the cell membranes:

a) Tight junction (occluding junction) (zero)

b) Adhering junction (anchoring junction) (20nm)

c) Gap junction (communicating junction) (2nm)

2- **According to the shape of the junction**:

a) Zonula junction (girdle or ring junction)

b) Fascia junction (band junction)

c) Macula junction (spot junction)

- **Matching of both types of classification will produce the following types of junctions (those consisting of both distance and shape together:**

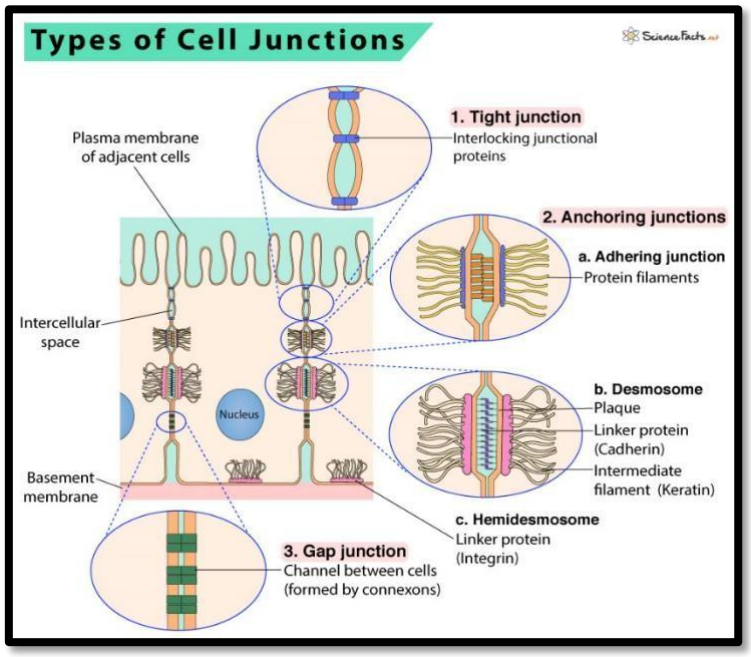
A. **Tight (occluding) junctions**:

▪ Zonula occludens. ▪ Fascia occludens.

B. **Adhering (anchoring) junctions**:

▪ Zonula adherens. ▪ Fascia adherens. ▪ Macula adherens.

C. **Communicating junctions (gap junctions)**.



Macula adherens (desmosomes):

- **Definition**: In this type of junction the distance is **15-20 nm** and the shape is localized small circular spot like area.
- **Structure**:

✓ Cytoplasmic aspect of the junctional membranes has dense areas called **plaques** formed of microfilaments. The plaque is **spot** like.

✓ **Intermediate filaments** are inserted into these plaques and loop back towards the cytoplasm.

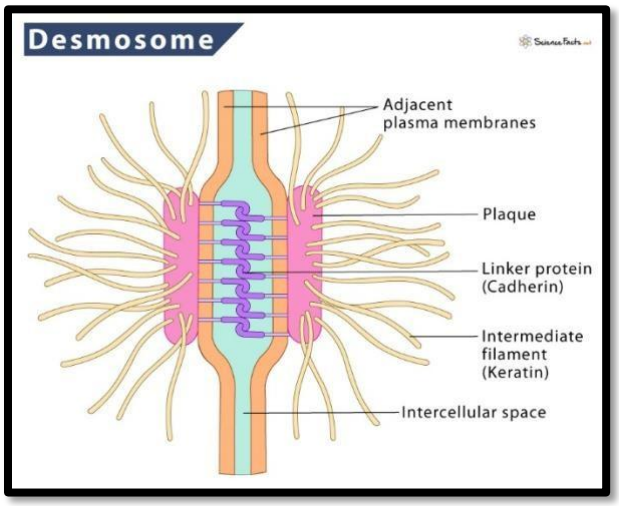
✓ Intercellular space contains a thin electron dense material.

- **Sites**:

1. Between cells of the epidermis of the **skin**.
2. Between **cardiac** muscle cells.
3. Between **intestinal** epithelial cells as a part of junction complex.

- **Function of Adhering Junctions**:

4. Provide strong attachment to the adjacent cells.
5. Prevent passages of materials through the intercellular space
6. Links the **cytoskeleton** of the two cells.



Cell Junction Complex:

- **Definition**: It means more than one type of cell junctions found with each other between certain cells at certain sites.
- **Example**:

✓ Junctional complex between upper parts of **intestinal** absorptive epithelial cells.

✓ It includes **zonula occludens**, **zonula adherens** and **macula adherens** (**desmosomes**).

Communicating Junctions (Gap Junctions or Nexuses):

- **Definition**: one of cell junction in which the distance is **2-3nm** and the shape is localized **small circular** and **spot** like
- **Structure**:

✓ At the site of junction, there are small structures called **connexons** (channel-forming proteins) that bridge the space between the two cells.

✓ Each connexon is formed of **six cylindrical subunits** (transmembrane proteins called **connexins**) arranged around a central channel.

✓ These connexons links the interiors of the two cells and allow ions and small molecules to pass from one cell to another without entering the intercellular space.

- **Sites**:

✓ Between **cardiac muscle** cells.

✓ Between **smooth muscle** cells.

✓ Between **epithelial** cells.

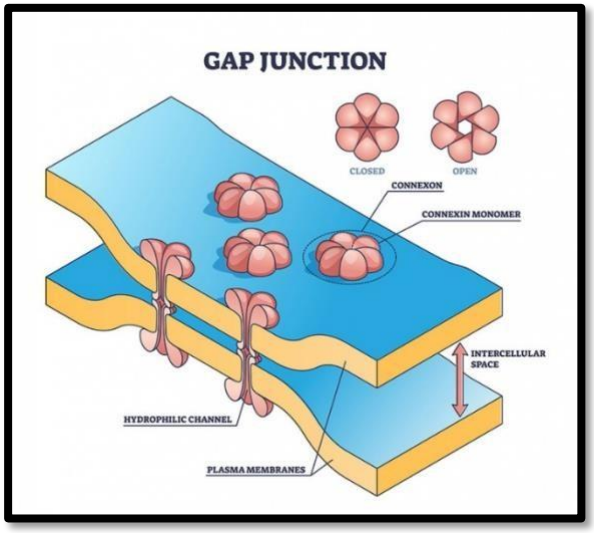
- **Function**:

1 Provide strong attachment to the adjacent cells.

2 Maintain direct communication between cells for transmission of ions, amino acids and hormones

3 **Allow spreading** of excitation waves throughout the heart and smooth muscles.

4 **Transmit impulse** at certain synapse.



C. **Basal surface specializations**:

- **The specific structures found on the basal surface include**:

a) Basement Membrane:

- **Site**: It is a supporting layer present between epithelium and the underlying connective tissue.

- **LM**:

✓ It **cannot** be seen by **H&E**.

✓ It needs special stains as **PAS**, **silver** or **trichrome stains**.

- **EM**:

1. Basal Lamina:

✓ It is secreted by the cells themselves.

✓ **It consists of**:

a) Lamina lucida:

✓ Adjacent to the base of epithelial cell membrane.

✓ Formed of **glycoproteins**.

✓ Appears electron **lucent**.

b) Lamina densa:

✓ Present adjacent to the underlying connective tissue.

✓ Formed of type **IV collagen** fibers.

✓ Appears electron **dense**.

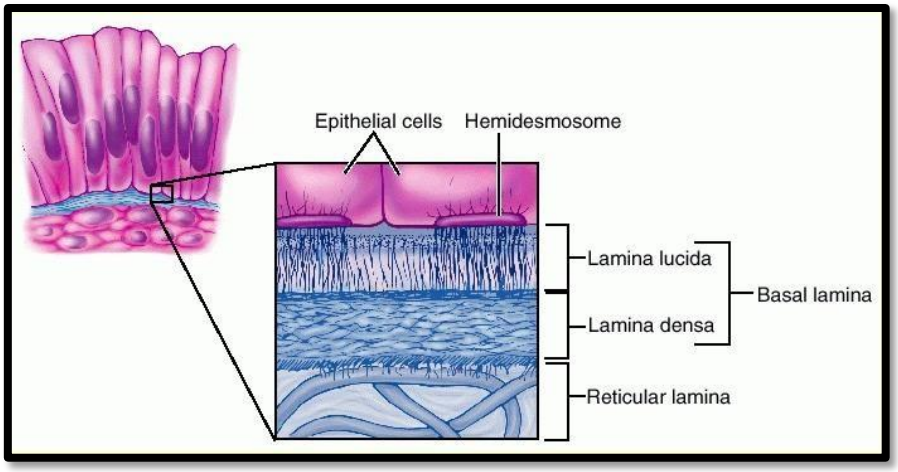
2. Reticular Lamina (lamina reticularis):

✓ Found between basal lamina and the underlying connective tissue.

✓ Synthesized by connective tissue cells (**fibroblasts**).

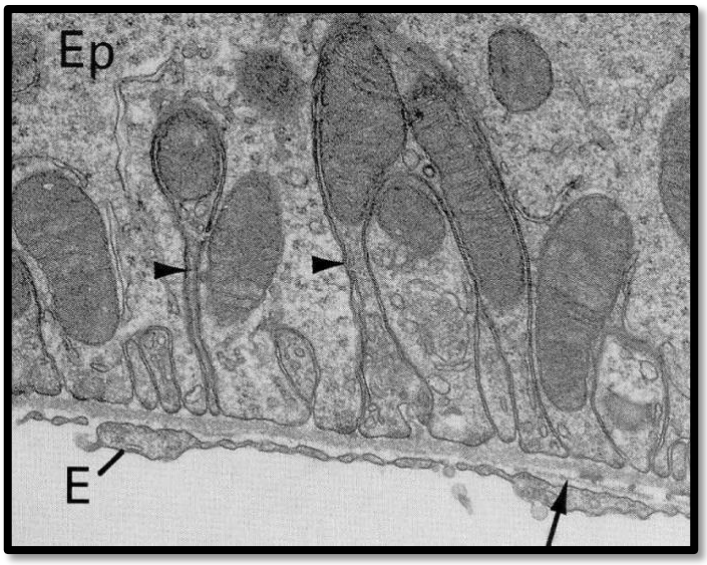
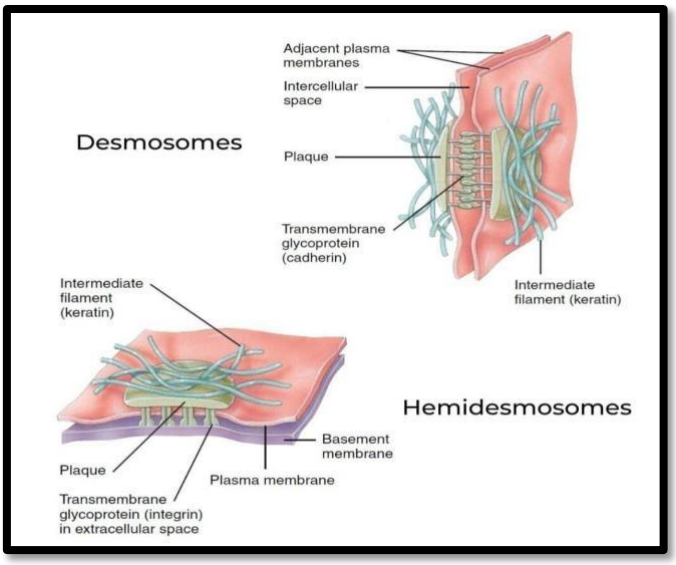
✓ It is formed of type **I** and **III collagen** fiber.

- **Function of basal lamina**:
  1. **Support** of the overlying epithelium.
  2. **Forms attachment** between epithelium and underlying connective tissue.
  3. **Prevents** passages of **large** molecules and **allows** passage of **small** molecules (i.e. filtration) as in the kidney.
  4. Regulate epithelial cell **proliferation** & guides their **migration** during repair after injury.



b) Basal Infoldings:

- **Definition**: They are **invaginations** of the basal cell membrane dividing the basal cytoplasm into several compartments. These cytoplasmic compartments contain **mitochondria** to provide energy needed for active transport.
- **Site and function**: They are present in the epithelial cells that transport fluid and have metabolic function (**ion-transporting epithelial cells**) for increasing surface area for transport. E.g. kidney tubules and striated ducts of salivary glands.

C) Hemidesmosomes:

- **Structure**: half (½) of desmosomes.
- **Site**: between bases of epithelial cells and basal lamina.
- **Function**: help to attach bases of epithelial cells to basal lamina.